CloudNine Systems

Group 4 | Milestone 1 ERD

30 tables organized into 6 connected modules.

Logical relational design in Third Normal Form (3NF).

Each module page shows all owned attributes and incoming FK relationships.

Gray entities are references to tables fully defined on another module page.

Relationship ends: 1 = required parent; 0..1 = optional parent;

0..* = zero or more children; 0..1 child = unique foreign key.

PK = primary key; FK = foreign key; UK = additional unique key.

? means nullable. Composite PKs are marked on every participating column.

Minimum child counts are enforced by lifecycle rules, not by ordinary FKs.

XOR, time interval, capacity and accounting rules appear in the analysis report.

All monetary values are USD. All timestamps are UTC.

The editable Mermaid and generation script accompany this PDF.

3. Clients and people

4. Infrastructure and history

5. Reservations and revisions

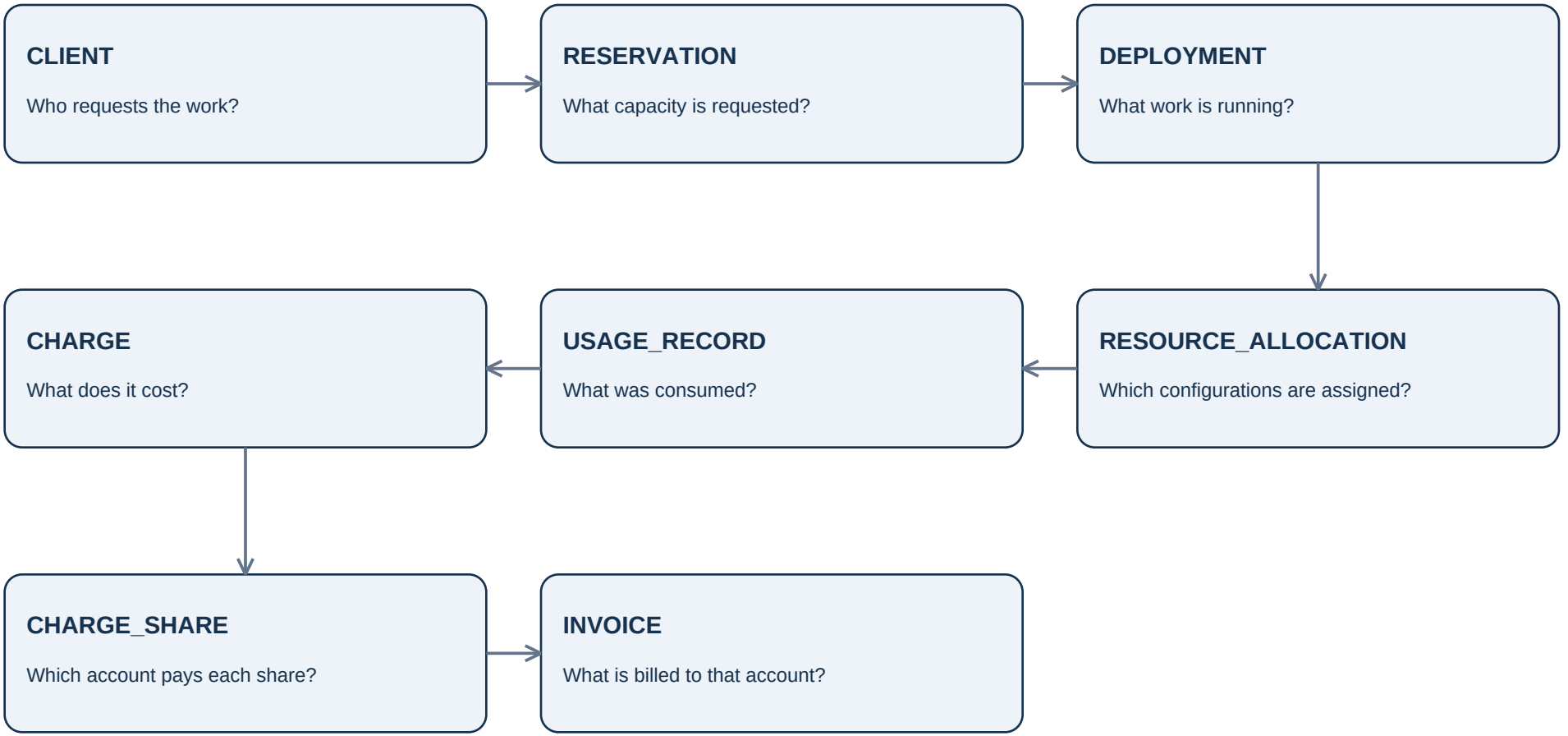
6. Deployments and assignments

7. Services, rates and metering

8. Billing and invoices

Core workflow overview

Read this first; detailed entities, keys and cardinalities follow on the module pages.

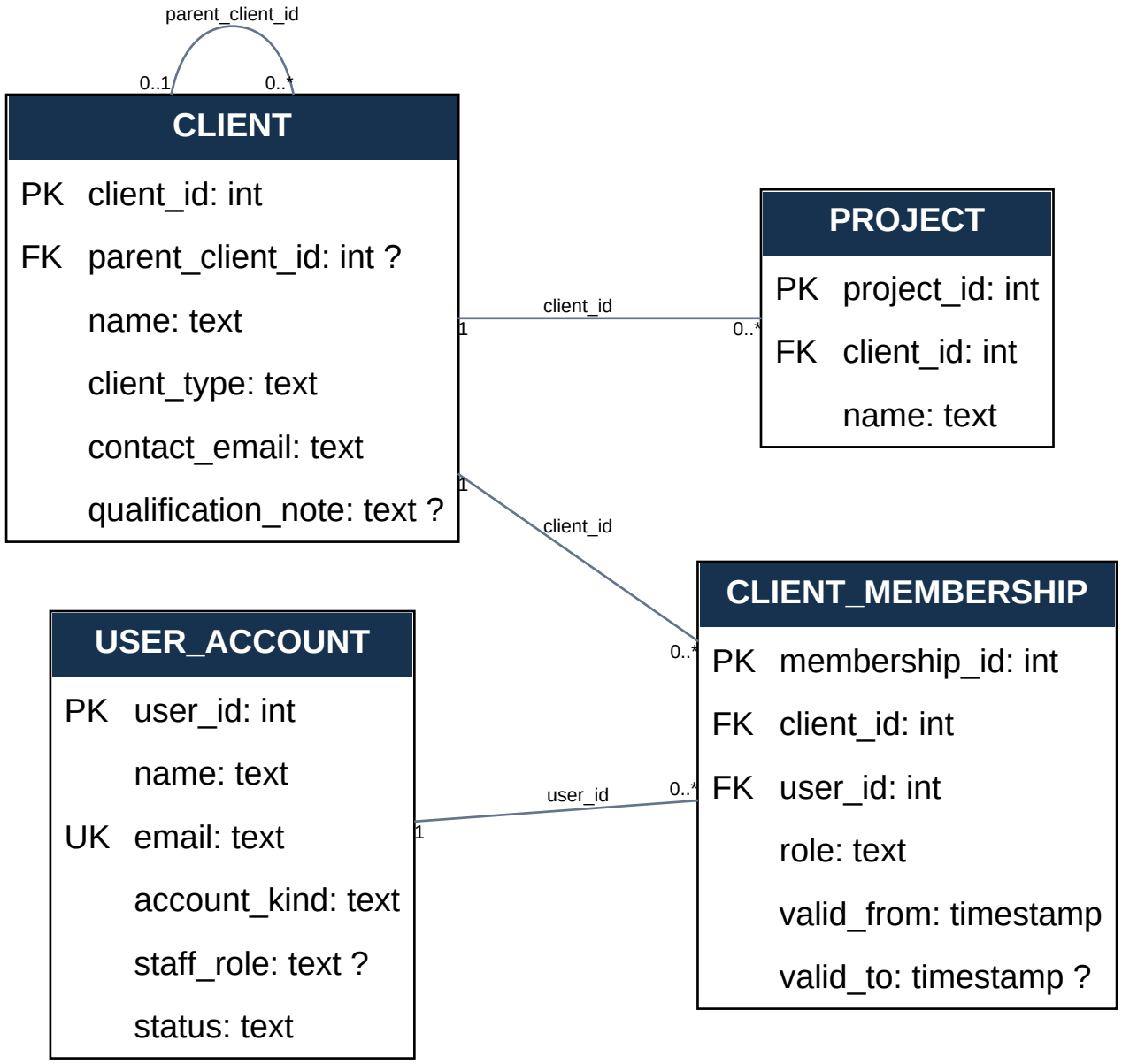


Conceptual flow only: intermediate service, pricing and invoice-line tables are omitted here.

The following six module pages define the full 30-table logical model and FK multiplicities.

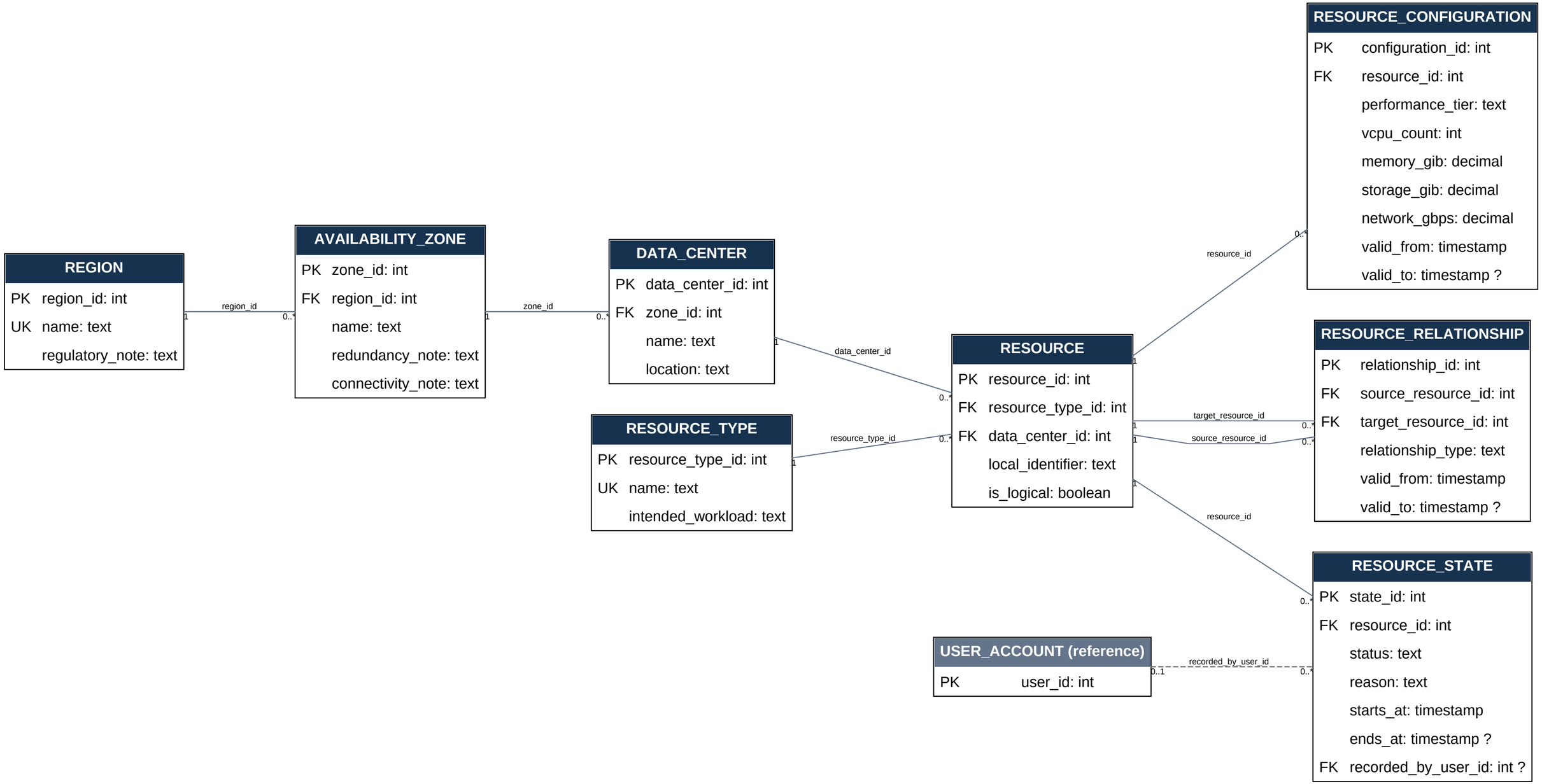
1. Clients and people

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



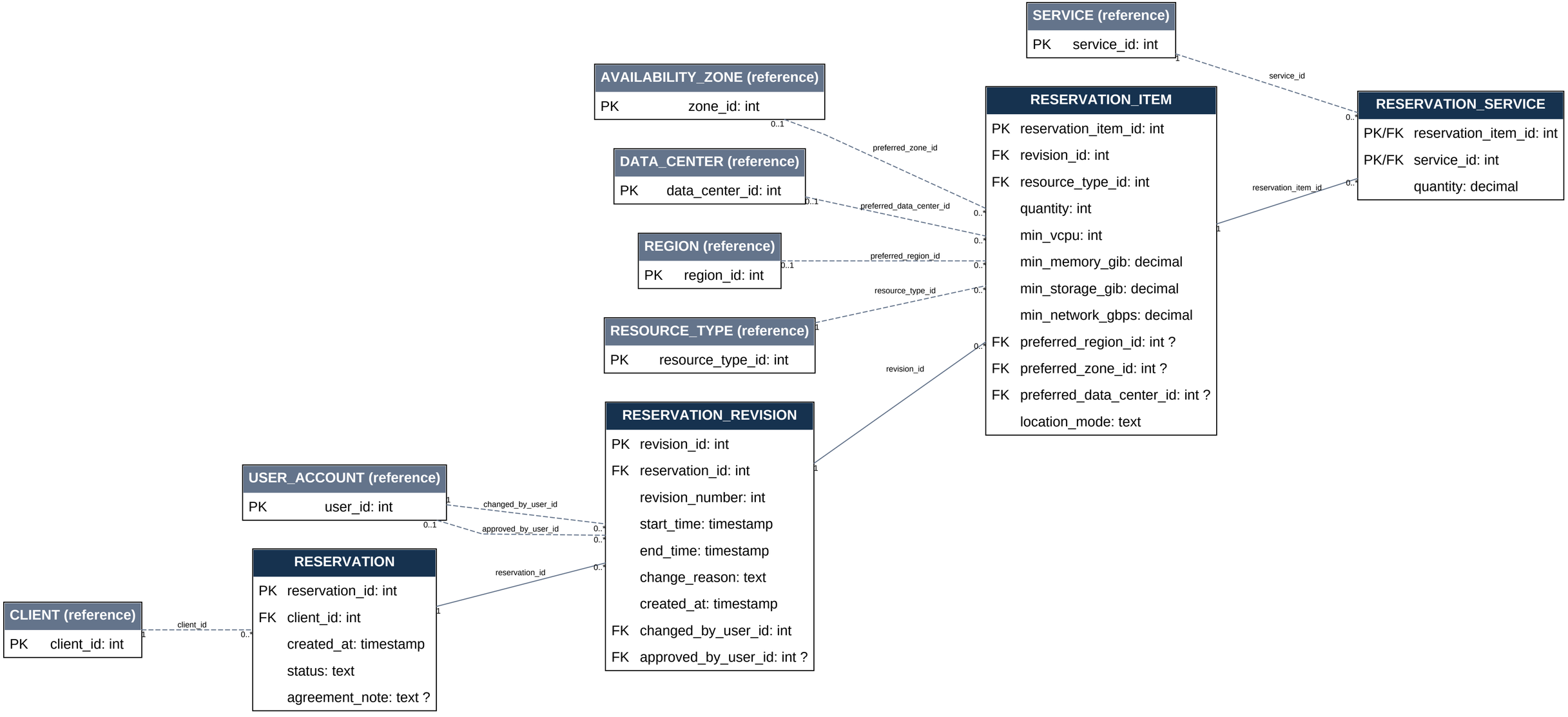
2. Infrastructure and history

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



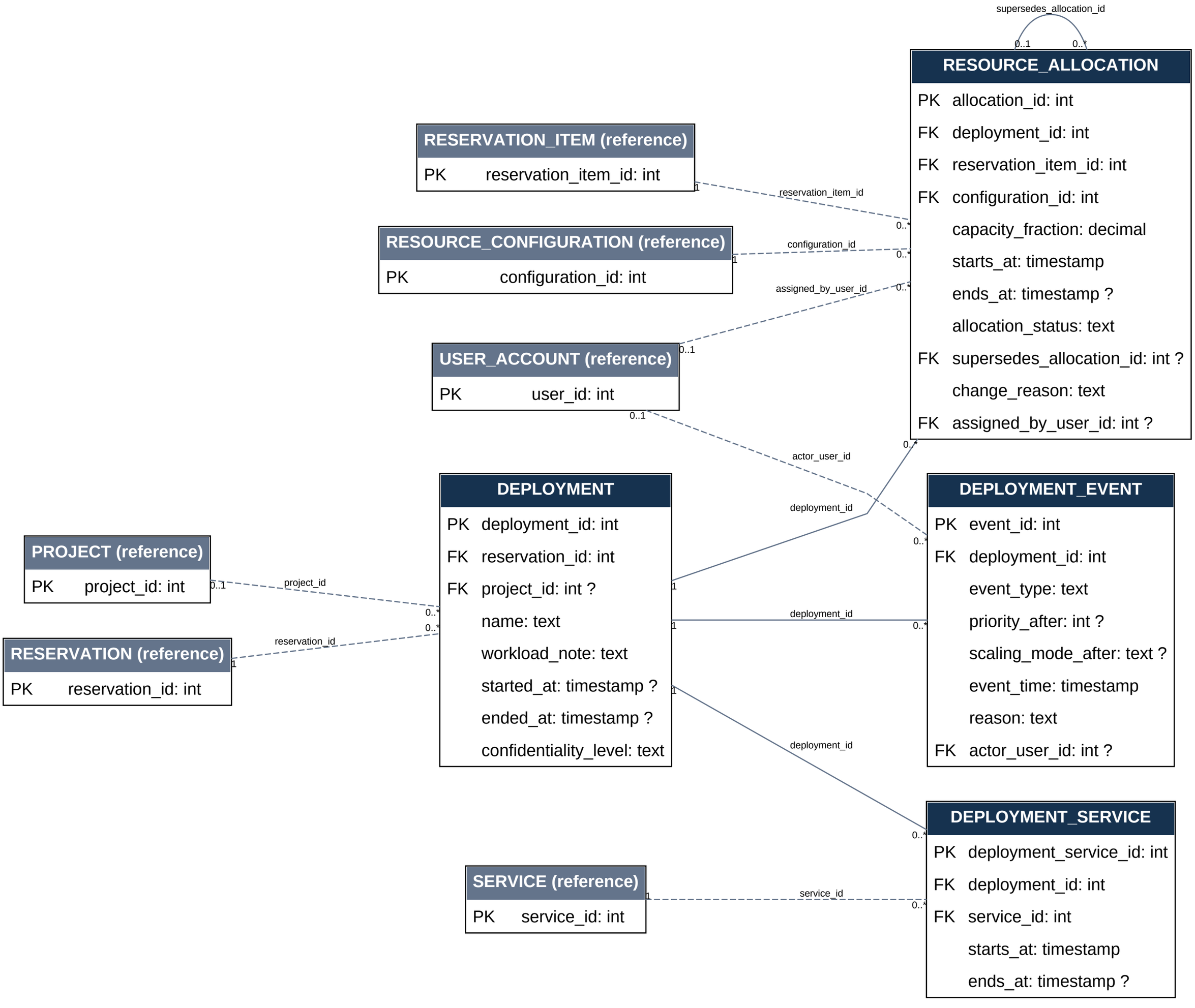
3. Reservations and revisions

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



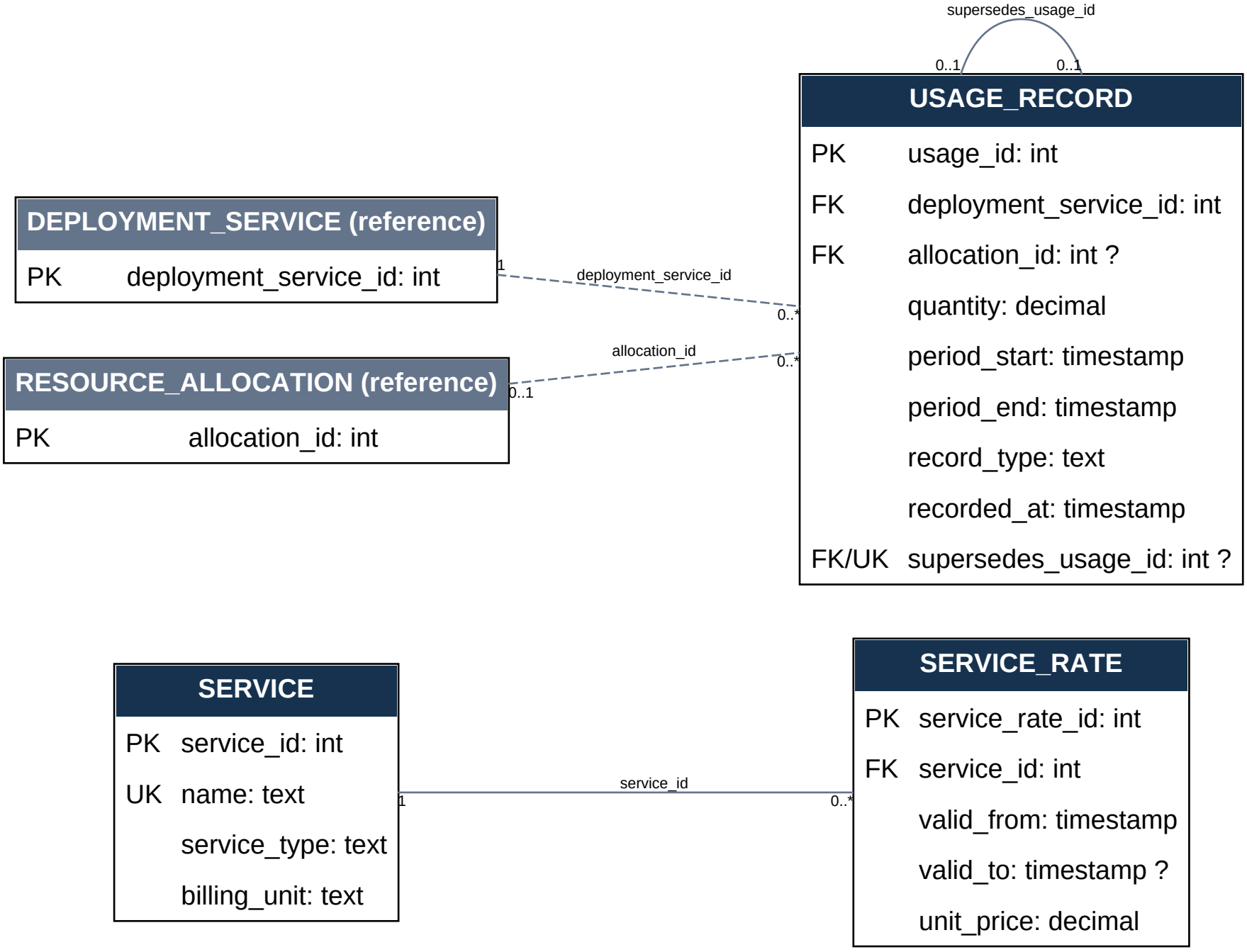
4. Deployments and assignments

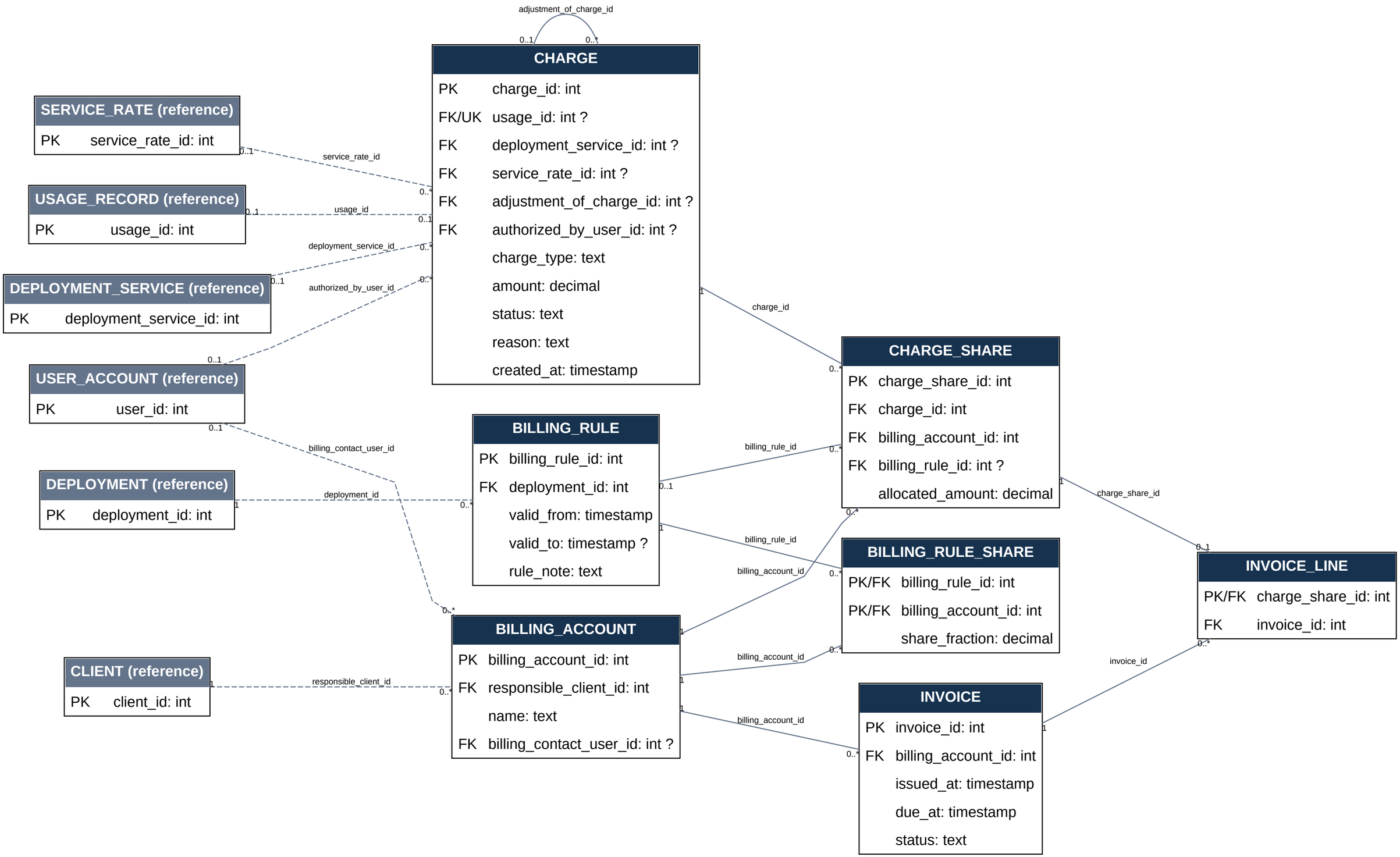
PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



5. Services, rates and metering

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end



6. Billing and invoices

PK / FK / UK | ? nullable | Gray = external reference | Edge labels give multiplicity at each end